**2** **(a)** Define velocity.

.................................................................................................................................

................................................................................................................................. [1]

**(b)** A student throws a ball over a vertical wall of height $h$, as shown in Fig. 2.1.

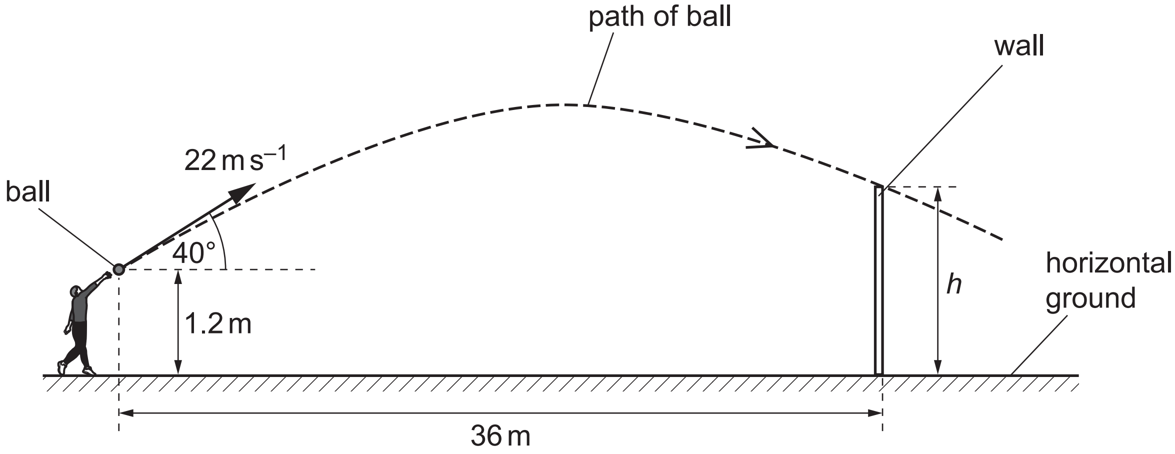


**Fig. 2.1** (not to scale)

The ball leaves the hand of the student at a height of 1.2 m above the horizontal ground.
The ball has an initial velocity of $22\,m\,s^{-1}$ at an angle of 40° to the horizontal.
The wall is a horizontal distance of 36 m from where the student releases the ball.

Air resistance is negligible.

**(i)** Determine the time taken for the ball to reach the wall.

time taken = ...................................................... s [2]

**(ii)** Calculate the vertical component $u$ of the initial velocity of the ball.

$u$ = .................................................. $m\,s^{-1}$ [1]

**(iii)** The ball just goes over the wall.

Calculate the height $h$ of the wall.

$h$ = ...................................................... m [3]

[Total: 7]